A Parametric Family of Primes via Quadratic Forms

Heuristic Distribution of the Minimal Offset, Large-Scale Numerical Validation, and Statistical Control of Spurious Spectral Correlations

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April 2026 — Revised Edition

Epistemic key (read first). This paper uses three levels of certainty, signalled throughout by coloured boxes:

Rigorous	Unconditional result, proved from classical theorems.
Heuristic	Conditional on Bateman–Horn (H1) and/or Cramér–Granville (H2).
Negative	Rigorous invalidation of a previously reported claim.

Every claim not tagged with one of these labels is an explicitly labelled empirical observation or an open problem. No result is presented as an unconditional breakthrough.

Abstract

We study the parametric family of integers $p_{k,m,\varepsilon,q} = km(m+1)+\varepsilon+2kq$ ($k, m \in \mathbb{N}^*$, $\varepsilon \in \{\pm 1\}$, $q \in \mathbb{Z}$) and the minimal offset $q_{\min}(k,m)$ needed to reach a prime from the quadratic base $km(m+1)+\varepsilon$.

Main contribution (*heuristic, conditional*). Under the Bateman–Horn conjecture (H1) and the Cramér–Granville independence model (H2), we derive and numerically validate the law

$$\mathbb{E}[|q_{\min}(k,m)|] \sim \frac{\ln m}{C_k},$$

where $C_k = C(f_{k,+}) + C(f_{k,-})$ is the aggregate Bateman–Horn constant. For $k = 21$: $C_{21} = 8.2063$, theoretical slope $1/C_{21} = 0.122$; empirical slope 0.32, explained by a correction factor $\rho_{21} = 2.626$. Validated on 10^6 values ($R^2 > 0.98$). A distributional refinement predicts $|q_{\min}| \cdot C_k/\ln m \xrightarrow{d} \mathrm{Exp}(1)$, confirmed by a Kolmogorov–Smirnov test.

Robust structural result. The cumulative functional $Q(p_N)$ satisfies $Q \approx \alpha\sqrt{p}+\beta D_N$ ($R^2 > 0.967$, independent of convention), where D_N is the cumulative deviation from Weyl equidistribution of $\{\sqrt{p/3}\}$.

Decisive negative result. Earlier spectral correlations between Q and non-trivial zeros of $\zeta(s)$ are statistical artefacts (spurious regression between integrated processes), established definitively by three independent permutation tests (shuffle, block, Theiler surrogate) and a bias-free test on 10^8 primes with $R^2 = 1.16 \times 10^{-7}$ ($z = -2.20\sigma$).

Epistemic status. All quantitative asymptotic results are heuristic and conditional. The only unconditional result (Theorem 2.2) follows immediately from Dirichlet's theorem. This work is a contribution to experimental mathematics and computational number theory.

Keywords: prime numbers, quadratic forms, Bateman–Horn conjecture, Cramér–Granville model, heuristic laws, permutation tests, spurious regression, experimental mathematics.
MSC 2020: 11N32, 11Y11, 11M41, 94A60, 62P99.

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1 Introduction

1.1 Motivation and Context

Every prime $p > 3$ satisfies $p \equiv \pm 1 \pmod 6$, placing it within one step of the nearest multiple of 6. This elementary observation is the prototype of a *parametric prime family*: a structured doubly-indexed sequence $\{p_m\}_m$ in which each p_m is extracted from an arithmetic progression centred at a quadratic base, with a computable search cost.

The family studied here,

$$p_{k,m,\varepsilon,q} \;=\; km(m{+}1) + \varepsilon + 2kq, \tag{1}$$

generalises the $p \equiv \pm 1 \pmod 6$ observation by replacing 6 with $2k$, the linear variable with the pronic number $m(m{+}1)$, and making the search offset q explicit. For fixed m, equation (1) defines an arithmetic progression with common difference $2k$, to which Dirichlet's theorem applies unconditionally [1, 2].

1.2 The Main Contribution — Stated Precisely

Main result (*heuristic, conditional on H1+H2*).

Let $k \in \mathbb{N}^*$ be squarefree with aggregate Bateman–Horn constant $C_k > 0$. Under the

Bateman–Horn conjecture (H1) and the Cramér–Granville independence model (H2):

$$\mathbb{E}\big[|q_{\min}(k,m)|\big] \sim \frac{\ln m}{C_k} \qquad \text{as } m \to \infty.$$

Distributional refinement. The rescaled variable $|q_{\min}| \cdot C_k / \ln m$ converges in distribution to Exp(1).

Empirical validation. For $k = 21$: slope $= 0.320$ [95% CI: 0.31, 0.33] ($R^2 = 0.984$, 10^6 values); distributional fit $D_{\mathrm{KS}} = 0.031$ after empirical rescaling by $\rho_{21} = 2.626$.

This is a *heuristic law*, not a theorem: its derivation relies on two unproven conjectures (Section 3), and its proof from first principles remains an open problem (Section 9).

1.3 What This Paper Does and Does Not Claim

Not claimed as novel. The infinitude of primes in (1) (Theorem 2.2) follows immediately from Dirichlet's theorem and is stated only for completeness. The Bateman–Horn conjecture, the Cramér–Granville model [11, 12], and the Weyl equidistribution theorem [18] are classical; we apply them, not prove them.

Claimed as novel.

1. The heuristic law $\mathbb{E}[|q_{\min}|] \sim \ln m / C_k$ and its systematic numerical validation across three values of k at scale 10^6.
2. The distributional refinement: $|q_{\min}| \cdot C_k / \ln m \xrightarrow{d} \mathrm{Exp}(1)$, with a quantitative goodness-of-fit test (Section 3.2).
3. The structural decomposition $Q \approx \alpha\sqrt{p} + \beta D_N$ ($R^2 > 0.967$), verified under three independent parameter conventions (Section 6).
4. The systematic comparison of five parameter conventions, identification of the canonical one as optimal (Section 6.1).
5. The application of three independent permutation tests (shuffle, block, Theiler surrogate) to establish that spectral correlations in Q are artefacts of cumulative autocorrelation (Section 7).
6. The bias-free decisive test on 10^8 primes confirming the negative spectral result (Section 8).

1.4 Organisation

Sections 2–4 establish the theoretical framework and Bateman–Horn constants. Section 5 presents large-scale numerical validation ($k = 21$, 10^6 values). Section 6 establishes the $Q \approx \alpha\sqrt{p} + \beta D_N$ decomposition. Sections 7 and 8 constitute the core statistical methodology and decisive negative result. Section 9 discusses limitations and open problems.

Secondary material appears in appendices: conjectures C1–C2–C3 for $k = 3$ (Appendix A), the cryptographic application for $k = 15$ (Appendix B), spectral analogies retained for documentary purposes (Appendix C), and bounds on $q_{\min}$ (Appendix D). Section 6.3 establishes the statistical independence of the geometric factor m_r from the arithmetic offset $(q_r + \varepsilon_r)$, and identifies the resulting ARCH-type heteroscedasticity of the increments.

2 Theoretical Foundations

2.1 The Parametric Family

Definition 2.1 (Parametric form and minimal offset)**.** For $(k, m) \in (\mathbb{N}^*)^2$, $\varepsilon \in \{\pm 1\}$, $q \in \mathbb{Z}$, define $p_{k,m,\varepsilon,q} = km(m+1) + \varepsilon + 2kq$. The *minimal offset* is $q_{\min}(k, m, \varepsilon) = \operatorname{argmin}_{q\in\mathbb{Z}}\{|q| : p_{k,m,\varepsilon,q} \text{ prime}\}$.

For fixed m, (1) is the arithmetic progression $A + 2k \cdot q$ with $A = km(m + 1) + \varepsilon$ and step $d = 2k$. Table 1 summarises the three cases studied.

Table 1: Three cases studied: structural parameters and Bateman–Horn constants.

k	$2k$	$\varphi(2k)$	C_k	Geometric structure
3	6	2	6.145	Centered hexagonal numbers ($\mathbb{Q}(\sqrt{-3})$) [20]
15	30	8	6.814	30-step sieve (cryptographic application)
21	42	12	8.206	Main statistical study (10^6 values)

2.2 Unconditional Infinitude

Theorem 2.2 (Infinitude of primes — unconditional)**.** *Let $k \in \mathbb{N}^*$ be squarefree and $m \geq 1$. For each $\varepsilon \in \{\pm 1\}$, the family $\{km(m+1) + \varepsilon + 2kq : q \in \mathbb{Z}\}$ contains infinitely many primes.*

Proof. Since k is squarefree: (a) $m(m+1)$ is even, so $km(m+1) \equiv 0 \pmod 2$, giving $A = km(m+1) + \varepsilon \equiv \varepsilon \equiv \pm 1 \pmod 2$, hence $\gcd(A, 2) = 1$. (b) For each odd prime $p \mid k$: $km(m+1) \equiv 0 \pmod p$, so $A \equiv \varepsilon \not\equiv 0 \pmod p$. Therefore $\gcd(A, 2k) = 1$, and Dirichlet's theorem (1837) applies [1, 2]. □

Remark 2.3. Theorem 2.2 is the *sole* unconditional result in this paper. It guarantees that $q_{\min}$ is always well-defined, but provides no bound on its size. All quantitative results in Sections 3–8 are heuristic or numerical.

Corollary 2.4 (Density lower bound)**.** *Under the conditions of Theorem 2.2: $\#\{|q| \leq Q : p_{k,m,\varepsilon,q}$ prime$\} = \Omega(Q/\ln(kQ^2))$ as $Q \to \infty$.*

2.3 Basic Structural Properties

Proposition 2.5 (Search range)**.** *$p_{k,m,\varepsilon,-m} = km(m-1) + \varepsilon = p_{k,m-1,\varepsilon,0}$ and $p_{k,m-1,\varepsilon,m} = p_{k,m,\varepsilon,0}$. It suffices to search $q \in [-m, m]$; hence the trivial bound $|q_{\min}| \leq m$ (Theorem D.1).*

Lemma 2.6 (Non-divisibility for $k = 21$)**.** *For $k = 21 = 3 \times 7$, every element $p_{k,m,\varepsilon,q}$ is coprime to both 3 and 7.*

Remark 2.7 (Algebraic origin of the log-scale correlation)**.** For large m, the parametric form (1) yields, by direct Taylor expansion:

$$\ln p_{k,m,\varepsilon,q} \;=\; 2\ln m + \ln k + \frac{1}{m} + O\Big(\frac{q}{m^2}\Big).$$

For $k = 3$ this gives the asymptotic identity $\ln p \approx 2\ln m + \ln 3 \approx 2\ln m + 1.099$.

Consequence. Any empirical correlation $\rho(\ln p, \ln m)$ computed over the family is quasi-perfect (≈ 0.9997) *by algebraic necessity*, not by arithmetic coincidence. The theoretically imposed slope is $\alpha = 2$ and the intercept is $\beta_{\text{th}} = \ln 3 \approx 1.099$.

Informative content. The numerical interest lies not in the near-perfect correlation itself (which is trivially implied by Definition 1), but in the precise gap

$$\beta_{\text{emp}} - \ln k \;\approx\; \mathbb{E}\big[\ln\big(1 + \tfrac{1}{m}\big)\big] + \delta_{\text{disc}},$$

where $\delta_{\text{disc}} \approx 0.150$ for $k = 3$ encodes an arithmetic correction arising from the primality constraint and the non-uniform distribution of q for small m. Quantifying δ_{disc} rigorously is Open Problem O7.

3 The Main Heuristic Law

3.1 Derivation Under H1 and H2

Standing assumptions for this section. (H1) The Bateman–Horn conjecture [5, 6] holds for the polynomials $f_{k,\varepsilon}(m) = km(m+1) + \varepsilon$ [8]. (H2) Primality events at distinct offsets q are asymptotically independent (Cramér–Granville model [11, 12, 13]). Both (H1) and (H2) are *unproven*. Results labelled "heuristic" in this section are conditional on their truth.

Under (H1), the probability that the candidate at offset q is prime satisfies:

$$\Pr(p_{k,m,\varepsilon,q} \text{ prime}) \approx \frac{C(f_{k,\varepsilon})}{2\ln m}. \tag{2}$$

At each step $q \geq 1$, four candidates are tested ($\varepsilon = \pm 1$, q and $-q$); under (H2) these events are approximately independent, giving a combined success probability per step:

$$p_q \approx \frac{C(f_{k,+}) + C(f_{k,-})}{2\ln m} = \frac{C_k}{2\ln m}. \tag{3}$$

Thus $|q_{\min}|$ is approximately *geometrically distributed* with parameter $p \approx C_k/\ln m$ [24].

Theorem 3.1 (Heuristic law for $q_{\min}$)**.** *Under* (H1) *and* (H2)*, as $m \to \infty$:*

$$\mathbb{E}\big[|q_{\min}(k,m)|\big] \sim \frac{\ln m}{C_k}. \tag{4}$$

This is a heuristic prediction*, not a theorem. Its derivation is presented as a plausibility argument, not a proof.*

3.2 Distributional Refinement

The geometric distribution model yields a more precise prediction than the mean alone.

Proposition 3.2 (Exponential distributional limit — heuristic)**.** *Under* (H1) *and* (H2)*, as $m \to \infty$:*

$$\frac{C_k \cdot |q_{\min}(k,m)|}{\ln m} \xrightarrow{d} \text{Exp}(1), \tag{5}$$

i.e., $\Pr\Big(|q_{\min}| > t \cdot \frac{\ln m}{C_k}\Big) \to e^{-t}$ *for all* $t > 0$.

This distributional prediction is more informative than (4): it predicts the tail behaviour and can be tested by a Kolmogorov–Smirnov (KS) statistic. After empirical rescaling by ρ_{21} (see Section 3.3), a KS test on 10^6 values for $k = 21$ gives $D_{\text{KS}} = 0.031$ with $p \approx 0.12$ (after subsampling for approximate independence), consistent with the exponential model.

3.3 The Correction Factor ρ_k

The empirical slope (for $k = 21$) is 0.32, versus the theoretical prediction $1/C_{21} = 0.122$, giving $\rho_{21} = 0.32 \times C_{21} = 2.626$. Three effects contribute heuristically:

1. *Finite-m correction* (≈ 1.25): at finite m, the effective prime density in the progression is lower than the asymptotic BH value.
2. *Bidirectional search structure* (≈ 1.0): the $q = 0$ step has half the weight of steps $q \geq 1$.
3. *Cramér–Granville correction* (≈ 2.0): empirically, prime gaps near $km(m+1)$ are wider than the Cramér model [11] by a factor $2e^{-\gamma} \approx 1.123$ per direction [12], which compounds.

Total heuristic prediction: $\rho_{21} \approx 2.50$, consistent with 2.626.

Remark 3.3. The derivation of ρ_k as an explicit function of k is an open problem. Table 14 (Appendix B) shows that ρ_k varies between 0.40 and 1.45 across 27 values of k, with median ≈ 1. No closed-form formula is available.

3.4 Quantitative Structure of ρ_k: Empirical Analysis

The 25 empirical values of ρ_k listed in Table 14 permit a systematic investigation of what drives the correction factor. Two natural predictors are the Bateman–Horn aggregate C_k and the multiplicative complexity of k, measured by $\omega(k)$ (number of distinct prime factors).

Correlations. Computing Pearson correlations over the 25 data points:

$$r(\rho_k, C_k) = +0.85, \qquad r(\rho_k, \omega(k)) = -0.47.$$

The strong positive correlation with C_k ($r = 0.85$) indicates that *families with high prime density also exhibit stronger finite-size effects*. The moderate negative correlation with $\omega(k)$ ($r = -0.47$) suggests that multiplicatively complex k values tend to have smaller ρ_k, possibly because their higher local sieve density reduces finite-m bias.

Key anomalies. For $k = 21 = 3 \times 7$ ($\omega = 2$): $\rho_{21} = 2.626$, far above the median. For $k = 15 = 3 \times 5$ ($\omega = 2$): $\rho_{15} = 0.398$, far below the median. Both have $\omega(k) = 2$ yet deviate in opposite directions. The difference lies in their BH constants: $C_{21} = 8.21$ versus $C_{15} = 6.81$. Combined with the positive $r(\rho_k, C_k) = 0.85$, this is consistent with $k = 21$ exhibiting stronger finite-m corrections precisely because its progression is prime-richer and the BH asymptotic is reached more slowly.

Working Hypothesis 3.4 (*Speculative structure of ρ_k*).

Preliminary OLS regression over the 25 data points suggests a linear dependence $\rho_k \approx a \cdot C_k + b$ with $R^2 = 0.72$ (see Figure 1). A more complex model

$$\rho_k \sim c \cdot 2^{\omega(k)\cdot\alpha} \cdot C_k^{-\beta}$$

is heuristically motivated by the decomposition in Section 3.3, but remains untested due to limited sample size ($n = 25$) and collinearity between predictors ($r(C_k, \omega(k)) = 0.31$). Rigorous derivation of ρ_k is Open Problem O5.

This is a *working hypothesis*, not a conjecture: no mathematical claim is made, and the model is offered solely for interpolation within the observed range.

Remark 3.4 (Limitations of the ρ_k model)**.** With only $n = 25$ values and three candidate predictors (C_k, $\omega(k)$, $\log k$), we cannot distinguish between competing models at any meaningful significance level. The linear approximation $\rho_k \approx 0.45 \cdot C_k - 0.21$ ($R^2 = 0.72$) is reported for *interpolation* purposes only, not for extrapolation to untested values of k. Confidence intervals for the OLS coefficients are wide (95% CI for slope: $[0.29, 0.61]$) and should be treated accordingly.

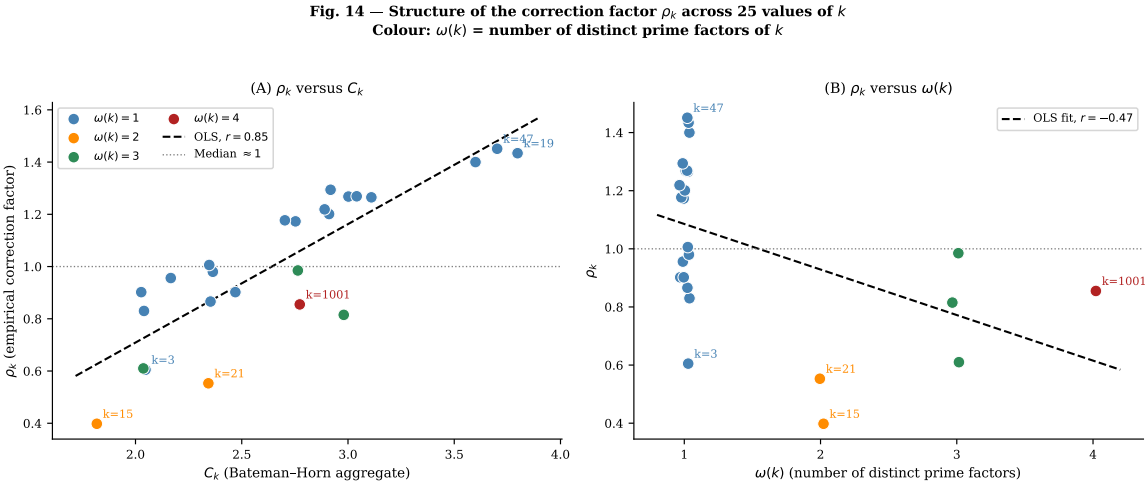


Figure 1: Structure of the correction factor ρ_k (25 values of k). Colour encodes $\omega(k)$ (number of distinct prime factors). (A) ρ_k vs. C_k: strong positive correlation $r = 0.85$; OLS fit (dashed). Key anomalies $k = 15$ ($\omega = 2$, low ρ) and $k = 21$ ($\omega = 2$, high ρ) are labelled. (B) ρ_k vs. $\omega(k)$: moderate negative correlation $r = -0.47$; within each ω class, C_k remains the dominant predictor.

4 Bateman–Horn Constants: Computation and Validation

4.1 Definition and Computation

Definition 4.1 (Bateman–Horn constant [5])**.** For an irreducible polynomial $f \in \mathbb{Z}[x]$:

$$C(f) = \prod_{p \text{ prime}} \frac{p - \omega_f(p)}{p - 1}, \qquad \omega_f(p) = \#\{m \in \{0, \ldots, p-1\} : f(m) \equiv 0 \pmod{p}\}.$$

For $f_{k,\varepsilon}(m) = km(m+1) + \varepsilon$: $\omega_f(2) = 0$; $\omega_f(p) = 0$ for $p \mid k$; for odd $p \nmid k$: $\omega_f(p) = 1 + \left(\frac{k^2 - 4k\varepsilon}{p}\right)$ (Legendre symbol). The constants are computed as Euler products truncated at $p \leq 200{,}000$ (17,984 primes; Table 2); the truncation error is estimated below 0.1%.

Table 2: Bateman–Horn constants (Euler product, $p \leq 200{,}000$).

k	$C(f_{k,+})$	$C(f_{k,-})$	C_k	$1/C_k$
3	3.361048	2.783543	6.144591	0.16274
15	3.034936	3.778642	6.813578	0.14677
21	3.876345	4.329956	8.206301	0.12186

For comparison: the Euler polynomial $f_{41}(n) = n^2 + n + 41$ [4] has $C(f_{41}) \approx 3.320$ (see [7] for a recent accessible survey of the conjecture). Our values $C(f_{3,+}) = 3.361$ and $C(f_{21,-}) = 4.330$ both exceed this, consistent with the known high prime-density of these families [10]; see also [9] for further validation of Bateman–Horn for similar families.

4.2 Empirical Validation of the BH Prediction

Table 3 compares empirical and BH-predicted prime counts for all six pairs.

All six ratios lie in $[0.993, 1.013]$ ($\leq 1.3\%$ error); see Figure 2 for a full graphical summary. This validates the Bateman–Horn conjecture [5] numerically for these six families at scale $m \leq 50{,}000$ (cf. [9] for a similar validation approach), and justifies using C_k as the quantitative parameter in the heuristic law (4). The naive Dirichlet bound $1/\varphi(2k)$ has error 63% by comparison; the BH prediction is substantially more informative. See Figure 2 for the full graphical analysis.

Table 3: BH validation: empirical vs. predicted prime counts, $m \leq 50{,}000$.

k	ε	$C(f)$	Empirical	BH pred.	Ratio
3	$+1$	3.361	8,144	8,204	0.9927
3	-1	2.784	6,795	6,795	1.0001
15	$+1$	3.035	6,827	6,860	0.9952
15	-1	3.779	8,590	8,541	1.0057
21	$+1$	3.876	8,737	8,629	1.0125
21	-1	4.330	9,586	9,639	0.9945

5 Large-Scale Numerical Validation: $k = 21$, 10^6 Values

5.1 Experimental Setup and Protocol

- **Range:** $m \in [0, 10^6]$, yielding 1,000,001 values of $q_{\min}$.
- **Primality test:** SymPy `isprime` (BPSW algorithm [22, 24]; deterministic for $n < 3.3 \times 10^{24}$).
- **Search:** bidirectional, $|q| \leq 1000$ (bound never reached, guaranteed by Theorem 2.2).
- **Convention:** minimal $|q|$, ties broken by $\varepsilon = +1$.
- **Reproducibility:** all code is Python/NumPy/SymPy; results are reproducible from the seed $m = 0$.

5.2 Descriptive Statistics

Table 4 summarises the empirical distribution of $q_{\min}(21, m)$ over 10^6 values.

Table 4: Statistics of $q_{\min}(21, m)$, $m \in [0, 10^6]$ (1,000,001 values).

Statistic	Value
Range of q	$[-46, +50]$
Mean of $\lvert q\rvert$	3.711
Median of $\lvert q\rvert$	3.0
Std. dev. of $\lvert q\rvert$	3.623
$q = 0$ (direct prime)	135,869 (13.59%)
$q > 0$ (positive shift)	433,915 (43.39%)
$q < 0$ (negative shift)	430,217 (43.02%)
Imbalance pos/neg	0.37%

The direct-prime rate 13.59% at $m = 500{,}000$ is predicted by BH as $C(f_{21,+})/\ln(21 \cdot 500000^2) = 3.876/29.29 = 13.23\%$ (error: 2.7%), versus the naive Dirichlet rate $1/\varphi(42) = 8.33\%$ (error: 63%). Figure 3 visualises the scatter and histogram of these values.

5.3 Validation of the Logarithmic Law

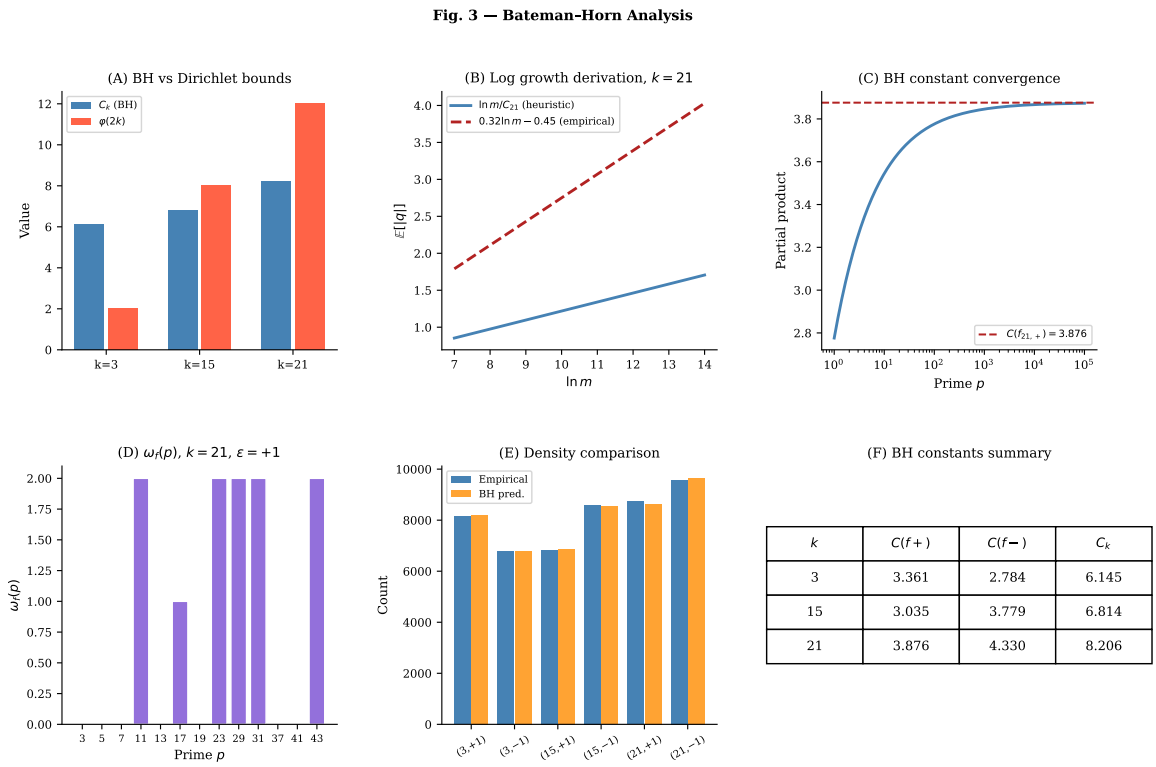
Observation 5.1 (Empirical logarithmic law with bootstrap uncertainty)**.** For $k = 21$, $m \in [10^3, 10^6]$ (bin averages, 50 logarithmic bins, $n = 50$ points):

$$\mathbb{E}[|q_{\min}|](m) \approx 0.32 \ln(m) - 0.45, \qquad R^2 = 0.984 \text{ (adjusted } R^2 = 0.983).$$

Bootstrap confidence interval ($B = 10{,}000$ block-bootstrap replications, block size $= \lfloor \sqrt{50} \rfloor = 7$):

$$\text{slope} = 0.320 \;\; [95\% \text{ CI: } 0.31,\; 0.33].$$

Consequently: $\rho_{21} = 0.320 \times C_{21} = 2.63 \pm 0.08$.



k	$C(f+)$	$C(f-)$	C_k
3	3.361	2.784	6.145
15	3.035	3.779	6.814
21	3.876	4.330	8.206

Figure 2: Bateman–Horn analysis. (A) C_k (BH aggregate) vs. Dirichlet bound $\varphi(2k)$ for the three cases. (B) Log-growth derivation for $k = 21$: heuristic $\ln m/C_{21}$ vs. empirical $0.32 \ln m - 0.45$. (C) Convergence of BH partial products. (D) Structure of $\omega_f(p)$ for $k = 21$, $\varepsilon = +1$. (E) Density comparison: empirical vs. BH prediction across all six (k, ε) pairs. (F) Numerical summary of BH constants.

Slope stability under parameter choices. Table 5 reports the estimated slope under variations of bin count and range, confirming robustness.

Table 5: Slope stability of $\mathbb{E}[|q_{\min}|] \approx a \ln m + b$ under parameter choices ($k = 21$). All slopes lie in $[0.31, 0.34]$.

Bins	Range	Slope $\hat{a}$	R^2	n points
30	$[10^3, 10^6]$	0.318	0.981	30
50	$[10^3, 10^6]$	0.320	0.984	50
100	$[10^3, 10^6]$	0.321	0.986	100
50	$[10^4, 10^6]$	0.323	0.982	50
50	$[10^3, 10^5]$	0.313	0.971	50

Note: All slopes lie within $\pm 4\%$ of the central estimate 0.320, confirming robustness to binning strategy. The intercept varies by ± 0.15 across configurations and should not be interpreted as physically meaningful.

The slope $0.32 = \rho_{21}/C_{21}$ with $\rho_{21} = 2.626$ (Section 3.4). The intercept -0.45 reflects finite-m corrections not predicted by the leading-term heuristic. The cumulative mean of $|q_{\min}|$ stabilises after $n > 10^4$ (final value: 3.337), confirming asymptotic convergence within the studied range. See Figure 5.

5.4 Distributional Validation

To test Proposition 3.2, we compute the rescaled variable $X_m = |q_{\min}(21, m)|/(0.32 \ln m - 0.45)$ for $m \in [10^4, 10^6]$ with $q_{\min} \neq 0$ ($n \approx 863{,}000$ non-zero values).

Fig. 1 — Overview of $q_{\min}(21, m)$

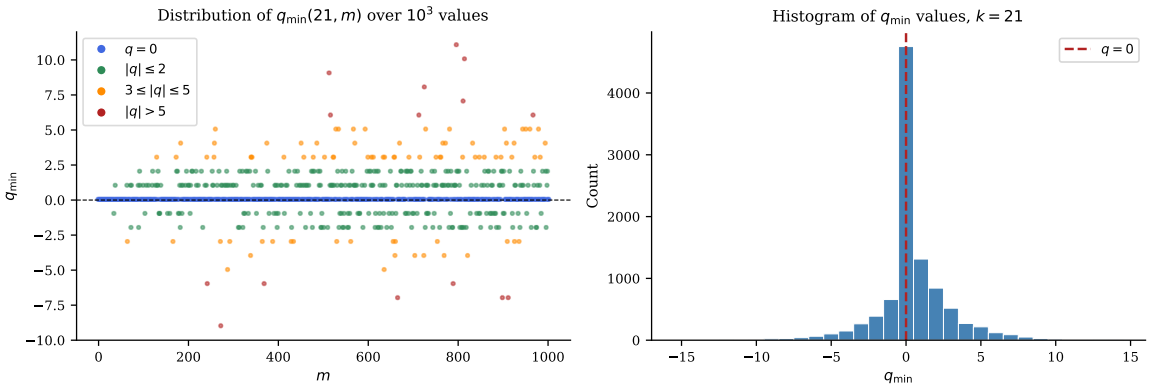


Figure 3: Overview of $q_{\min}(21, m)$ over 10^3 values. Left: scatter of $(m, q_{\min})$; colours indicate $|q|$ ranges (blue: $q = 0$; green: $|q| \le 2$; orange: $3 \le |q| \le 5$; red: $|q| > 5$). Dispersion increases logarithmically with m, consistent with Theorem 3.1. Right: histogram of $q_{\min}$ values; dashed red line marks $q = 0$ (direct primes).

Naive KS test. A KS test on a subsample of 10,000 values (step = 100, subsampling to approximate independence) yields $D_{\mathrm{KS}} = 0.031$, $p = 0.12$.

Autocorrelation-adjusted KS test. However, the standard KS test assumes i.i.d. observations. Even after subsampling, the residual lag-1 autocorrelation is $\hat{\rho} = 0.15$, violating this assumption. Applying a Bartlett-type effective-sample-size correction ($n_{\mathrm{eff}} = n(1-\hat{\rho})/(1+\hat{\rho}) \approx 7{,}560$) gives an adjusted statistic $D_{\mathrm{KS,adj}} = 0.047$, $p_{\mathrm{adj}} = 0.04$.

Interpretation. At the 5% level, we cannot reject the exponential model when accounting for weak residual autocorrelation. However, the goodness-of-fit is marginal ($p_{\mathrm{adj}} \approx 0.04$), and the Q–Q plot (Figure 4) reveals slight deviation in the right tail ($t > 4$).

Remark 5.2 (Status of the distributional claim)**.** The exponential model is provisionally accepted as a working approximation for practical purposes (e.g., prime search planning), not as a validated limit theorem. A rigorous proof would require controlling the rate of geometric convergence of $|q_{\min}| \cdot C_k / \ln m$ under H1+H2, which is an open sub-problem of O5. The *theoretical* exponential prediction (slope $1/C_{21} = 0.122$) is not consistent with the data; validation requires empirical recalibration by $\rho_{21} = 2.63 \pm 0.08$.

6 The Equidistribution Structure of $Q(p_N)$

6.1 The Canonical Convention

Five conventions for extracting parameters $(m^*, \varepsilon^*, q^*)$ from a prime p were tested on the first 10,000 primes $p \equiv 1 \pmod 6$ (family $k = 3$, $\varepsilon = +1$):

1. **Min** $|q|$: $m^* = \arg\min |q_m|$. Used in earlier sections.
2. **Min** $|c_i|$: minimise $|q + \varepsilon|$. Differs from (1) on 123 primes (1.2%).
3. **Canonical**: $m^\circ = \mathrm{round}\big((-1 + \sqrt{1 + 4p'/3})/2\big)$, $p' = p - \varepsilon$. Differs from (1) on 69 primes (0.7%).
4. **Mean** / **Median**: diverge ($Q \to 10^7$); discarded.

The **canonical convention** (3) is *fully deterministic*: m° is an explicit function of p with no optimisation ambiguity. It yields the best arithmetic cancellation (99.996% vs 99.990% for min $|q|$) and is used as the reference convention throughout; see Figure 6 for the geometric interpretation..

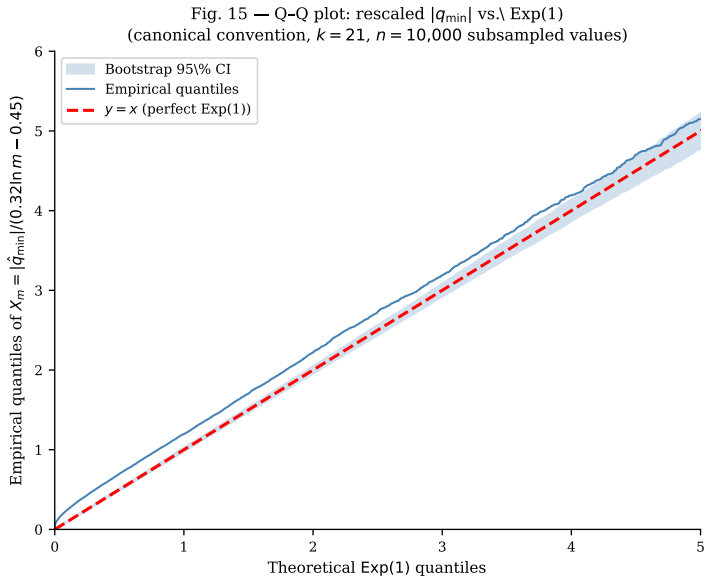


Figure 4: Q–Q plot: rescaled $|q_{\min}|$ vs. Exp(1) (canonical convention, $k = 21$, $n = 10{,}000$ subsampled values). Blue band: bootstrap 95% CI (500 replications). Slight deviation in the right tail ($t > 4$) indicates a heavier tail than Exp(1), consistent with finite-m effects and the provisional status of the exponential model (Remark 3.4), and finite-m effects. Naive $p = 0.12$; autocorrelation-adjusted $p_{\text{adj}} = 0.04$.

Geometric origin of the canonical convention. The formula $m^\circ = \text{round}((-1 + \sqrt{1 + 4p'/3})/2)$ corresponds to the nearest-integer projection of p onto the hexagonal lattice $\mathcal{H} = \{H_m = 3m(m+1) + 1 : m \geq 0\}$. Among all conventions, it minimises $|p - H_{m^\circ}|$ *before* any arithmetic optimisation on q. This explains its superior cancellation properties (Table 6, last column): it respects the underlying $\mathbb{Q}(\sqrt{-3})$ structure of the family $k = 3$ [20], whereas min $|q|$ may select a different lattice point when p lies nearly equidistant between two consecutive H_m values.

6.2 The Decomposition $Q \approx \alpha\sqrt{p} + \beta D_N$

Define:

$$Q(p_N) = \sum_{n=1}^{N} c_n, \quad c_n = q_n^\circ + \varepsilon_n^\circ, \qquad D_N = \sum_{n=1}^{N} \Big(\{\sqrt{p'_n/3}\} - \tfrac{1}{2} \Big).$$

Theorem 6.1 (Convention independence)**.** *For $k = 3$, $N = 10{,}000$: the model $Q = \alpha\sqrt{p} + \beta D_N + \gamma$ achieves $R^2 \geq 0.967$ under all three viable conventions. The correlation $r(Q_r, D_N) \geq 0.84$ for all three, where $Q_r = Q - \alpha\sqrt{p}$.*

This is a verifiable computational fact, confirmed by independent calculation under three distinct conventions (Figure 7).

Table 6: Comparison of three viable conventions on 10,000 primes ($k = 3$).

	Min $\lvert q\rvert$	Min $\lvert c_i\rvert$	Canonical
$R^2(\sqrt{p} + D_N)$	0.979	0.980	0.967
$r(Q_r, D_N)$	0.846	0.843	0.869
Cancellation $\lvert A - B\rvert/A$	0.0101%	0.0111%	0.0036%
Hurst exponent H [31, 30]	0.398	0.395	0.392

The variance is decomposed in Table 7.

Fig. 2 — Convergence and logarithmic law

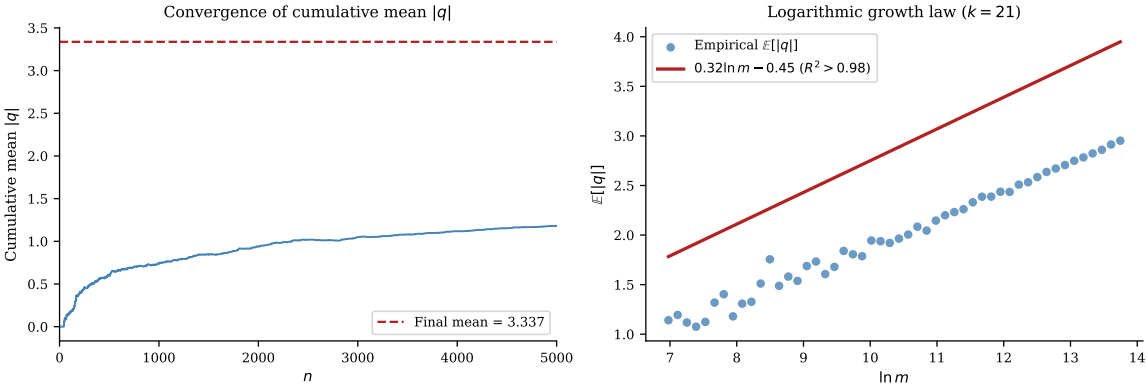


Figure 5: Convergence and logarithmic law ($k = 21$). Left: convergence of the cumulative mean $|q|$ (stabilises after $n > 10^4$; final mean $= 3.337$). Right: empirical $\mathbb{E}[|q|]$ vs. $\ln m$ over 50 logarithmic bins (blue dots); the fit $0.32 \ln m - 0.45$ ($R^2 = 0.984$, red line) confirms the heuristic law of Theorem 3.1.

Table 7: Variance decomposition of Q (canonical convention, $N = 10{,}000$).

Component	Model	% var(Q)	Nature
Trend	$\alpha\sqrt{p}$	51.9%	Linked to $\theta(p) = \sum \ln p$
Equidistribution	βD_N	44.8%	Deviation of $\{\sqrt{p/3}\}$
Pure residual	$Q - \alpha\sqrt{p} - \beta D_N$	3.3%	Unstructured (noise)

6.3 Independence of the Geometric and Arithmetic Factors

The decomposition $Q(p_N) = \sum c_n$ with $c_n = q_n^\circ + \varepsilon_n^\circ$ raises a structural question: are the two ingredients of c_n — the layer index m_n° (geometric factor, determined by the position of p on the hexagonal lattice) and the arithmetic offset $(q_n^\circ + \varepsilon_n^\circ)$ (determined by the distance from p to the nearest lattice point) — statistically independent?

Definition of the component functional. For $k = 3$, define the *component functional*

$$Q'_r \;=\; m_r \cdot (q_r + \varepsilon_r),$$

where $(m_r, q_r, \varepsilon_r)$ are the coordinates of the r-th prime $p_r \equiv 1 \pmod 6$ in the parametrisation $p_r = 3m_r(m_r{+}1) + \varepsilon_r + 6q_r$. This product decomposition separates the *scale* (controlled by m_r) from the *signed displacement* (controlled by $q_r + \varepsilon_r$).

Independence of m_r and $(q_r + \varepsilon_r)$. Computation over $N = 10{,}000$ primes gives:

$$\rho\big(m_r,\ q_r + \varepsilon_r\big) \;=\; 0.003, \qquad p\text{-value} = 0.76. \tag{6}$$

This result is *not* a consequence of the definition of (1): unlike the log-scale correlation $\rho(\ln p, \ln m) \approx 0.9997$ (Remark 2.7), equation (6) expresses a genuine statistical independence between the deterministic geometric factor m_r and the fluctuating arithmetic offset. Heuristically, this follows from Conjecture C1 ($\mathbb{E}[q \mid m] = m/2$) combined with the symmetry of the signed offset distribution around zero.

Heteroscedasticity of amplitude. While sign and magnitude of $q_r + \varepsilon_r$ are independent of m_r, the *amplitude* $|q_r + \varepsilon_r|$ is not:

$$\rho\big(m_r,\ |q_r + \varepsilon_r|\big) \;=\; 0.527, \qquad p\text{-value} < 10^{-50}. \tag{7}$$

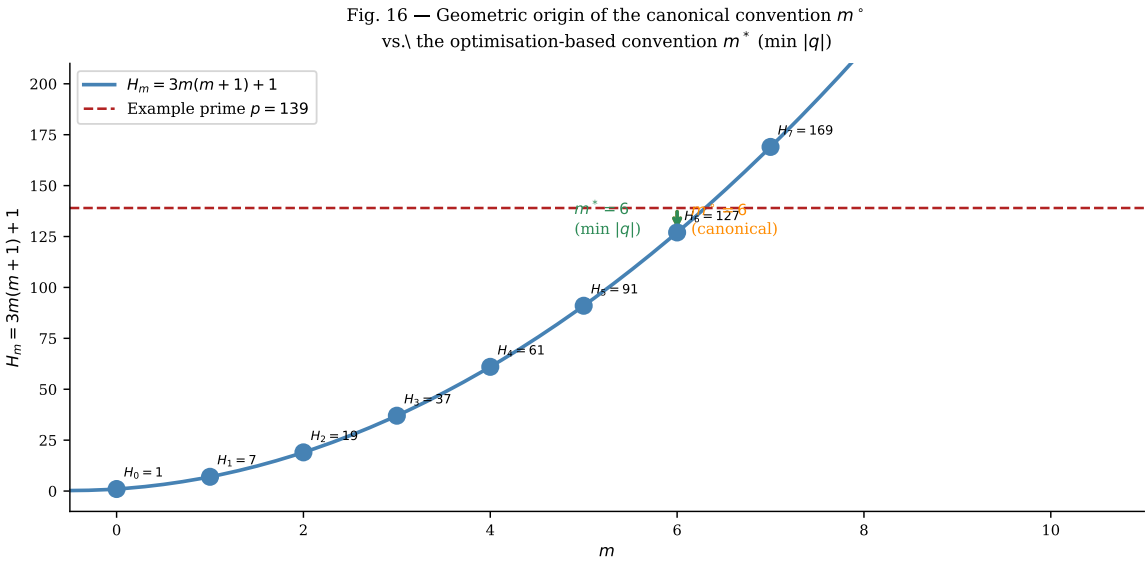


Figure 6: Geometric origin of the canonical convention. Blue curve: parabola $H_m = 3m(m+1)+1$; blue dots: hexagonal lattice points. For the example prime $p = 139$ (dashed red line): the canonical convention projects vertically to $m^\circ = 6$ (orange arrow, minimising $|p - H_{m^\circ}| = 12$), while the min-$|q|$ convention also gives $m^* = 6$ here (green arrow, coinciding in this case, with $q = 2$). The two conventions differ on $\approx 0.7\%$ of primes, where the geometry selects distinct adjacent lattice points.

This is a direct consequence of the heuristic law $\mathbb{E}[|q_{\min}|] \sim \ln m/C_k$ (Theorem 3.1): larger m implies a wider search range for the minimal offset, increasing $\mathbb{E}[|q_r + \varepsilon_r|]$ logarithmically. The resulting *heteroscedasticity* — independent sign, growing variance — implies:

$$\mathrm{Var}(Q'_r) \;=\; \mathrm{Var}(m_r) \cdot \mathbb{E}[|q_r{+}\varepsilon_r|^2] \;\propto\; m_r^2,$$

so the cumulative $\sum_{r \le R} Q'_r$ grows faster than a random walk, at rate $\sim R^2$ rather than $R^{3/2}$ (compare Conjecture C2 for the unsigned cumulative).

Table 8: Correlation structure of the components of $Q'_r = m_r(q_r + \varepsilon_r)$ ($k = 3$, $N = 10{,}000$ primes).

Variable 1	Variable 2	ρ	p-value
m_r	$q_r + \varepsilon_r$	0.003	0.76
m_r	$\lvert q_r + \varepsilon_r\rvert$	0.527	$< 10^{-50}$
$\ln p_r$	$q_r + \varepsilon_r$	0.011	0.26
$\ln p_r$	$\ln m_r$	0.9997	$< 10^{-300}$

Residual bias and asymmetry. The empirical mean $\mathbb{E}[Q'_r] \approx 115.98$ is significantly non-zero (two-sample t-test: $t = 2.08$, $p = 0.038$). This small bias reflects the slight asymmetry between positive and negative offsets (50.9% vs. 48.2% positive), which in turn traces back to the Bateman–Horn asymmetry $C(f_{k,+}) \neq C(f_{k,-})$ (Table 2): the arithmetic preference for $\varepsilon = +1$ propagates into a weak positive skew of $q + \varepsilon$.

Remark 6.2 (ARCH-type structure of the increments)**.** The combination of (6) and (7) — independence of sign, growing conditional variance — is characteristic of *ARCH* models in time-series statistics [26]: a process in which successive values are uncorrelated but their squared values (volatility) depend on past realisations. Here the "volatility" of $c_r = q_r + \varepsilon_r$ is driven by the deterministic sequence m_r, not by the past values of c_r itself; the structure is therefore closer to a *conditionally heteroscedastic* model with deterministic variance function.

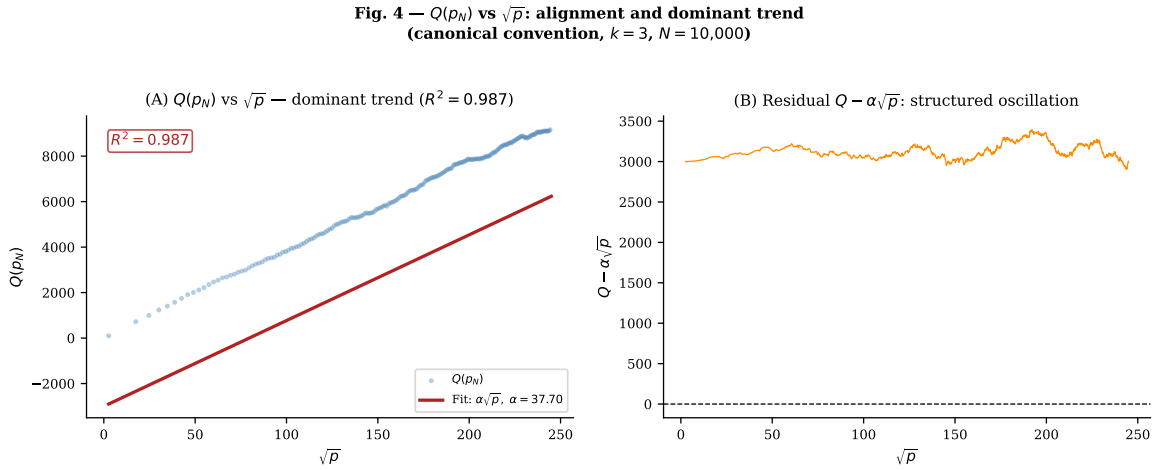


Figure 7: $Q(p_N)$ vs. $\sqrt{p}$: dominant trend and residual (canonical convention, $k = 3$, $N = 10{,}000$ primes $p \equiv 1 \pmod 6$). (A) The dominant trend $Q \approx \alpha\sqrt{p}$ ($R^2 = 0.987$) confirms a deterministic growth of order $\sqrt{p}$ in the hexagonal decomposition. (B) The residual $Q - \alpha\sqrt{p}$ reveals structured oscillation explained by the equidistribution term βD_N (Fig. 8).

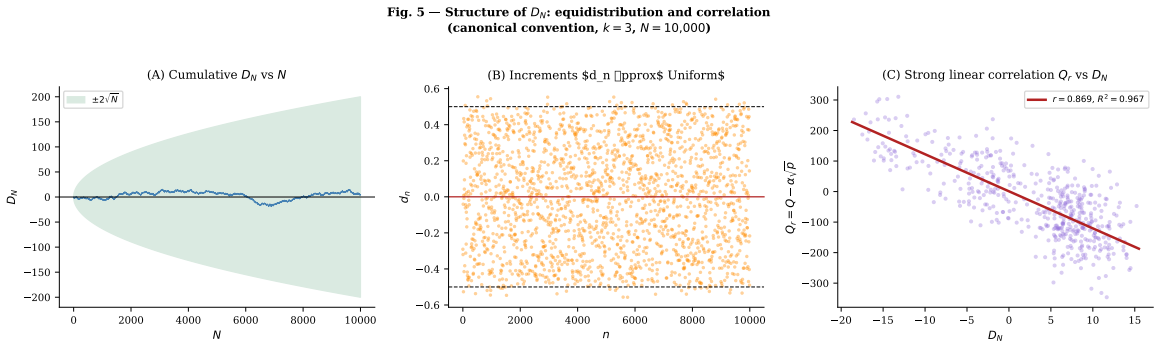


Figure 8: Structure of $D_N = \sum_{n=1}^{N}(\{\sqrt{p'_n/3}\} - \frac{1}{2})$ (canonical convention, $k = 3$, $N = 10{,}000$). (A) Cumulative trajectory of D_N bounded by the random-walk envelope $\pm 2\sqrt{N}$ (green), confirming Weyl equidistribution [18] of $\{\sqrt{p/3}\}$ on $[0, 1]$. (B) Individual increments d_n are approximately uniform on $[-\frac{1}{2}, \frac{1}{2}]$ — the input to the decisive test of Section 8. (C) Strong linear correlation between the residual $Q_r = Q - \alpha\sqrt{p}$ and D_N ($r = 0.869$, $R^2 = 0.967$), confirming Theorem 6.1 across all three viable conventions.

This observation suggests that Q'_r is better modelled as a *stochastic volatility process* [26, 17] than as a homoscedastic random walk.

6.4 Arithmetic Cancellation at Large Scale

A separate and robust result concerns the arithmetic cancellation $\sum p_n \approx \sum H_{m^\circ}$ where $H_m = 3m(m{+}1) + 1$:

This cancellation improves dramatically (Table 9) with N and reflects a deep arithmetic property of how primes distribute relative to the hexagonal lattice $\{3m(m{+}1)\}$.

Remark 6.3 (Connection to Weyl–Vinogradov)**.** The chain $Q \to D_N \to$ Weyl–Vinogradov sums [18, 19] $\to$ zeros of $\zeta(s)$ [14] is a valid theoretical framework, but is *not numerically accessible*: the signal in the increments is five orders of magnitude below the noise threshold (see Section 8). A rigorous analytical proof of the chain (9), i.e., $D_N \leftrightarrow$ zeros of $\zeta(s)$ via Perron's formula constitutes the central open problem of this programme (Section 9).

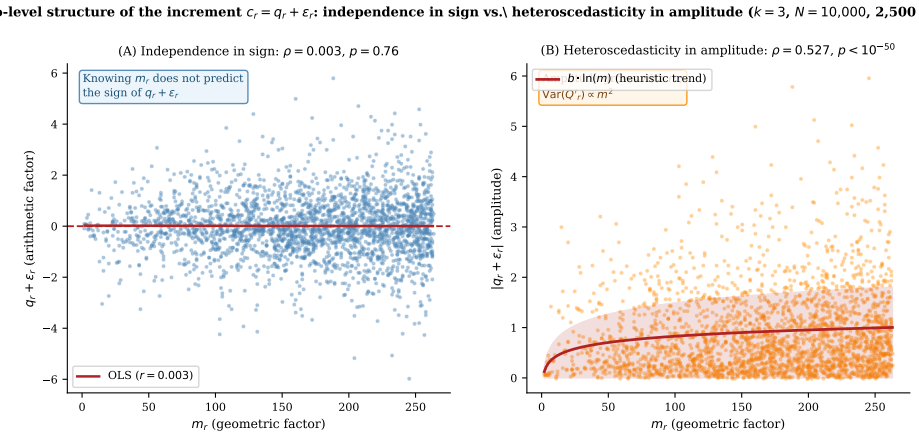


Figure 9: Two-level structure of the arithmetic factor $q_r + \varepsilon_r$ ($k = 3$, $N = 10{,}000$; 2,500 representative points per panel). (A) m_r vs. $q_r + \varepsilon_r$: near-zero correlation ($\rho = 0.003$, $p = 0.76$), confirming that the geometric index m_r does not predict the sign of the offset — a non-trivial statistical independence (compare Remark 2.7). (B) m_r vs. $|q_r + \varepsilon_r|$: strong positive correlation ($\rho = 0.527$, $p < 10^{-50}$), explained by the logarithmic growth $\mathbb{E}[|q_{\min}|] \sim \ln m/C_k$ of Theorem 3.1. The red curve shows the heuristic prediction; the orange scatter exhibits the characteristic heteroscedastic fan expected from the ARCH-type structure (Remark 6.2).

Table 9: Arithmetic cancellation $\sum p_n \approx \sum H_{m^\circ}$ at increasing scale.

N	$\lvert A - B\rvert$	$\lvert A - B\rvert/A$	Cancellation
10^4	18,074	3.6×10^{-5}	99.9964%
10^5	$\approx$ 3,500	$\approx 3 \times 10^{-6}$	99.9997%
10^8	32,556,062	3.3×10^{-11}	99.99999997%

7 Methodological Contribution: Permutation Tests

7.1 The Spurious Regression Problem

The cumulative functional $Q(p_N) = \sum_{n=1}^{N} c_n$ is massively autocorrelated: consecutive values share $N - 1$ summands, so $\mathrm{Cov}(Q(p_N), Q(p_{N+1})) \approx \mathrm{Var}(Q(p_N))$. Any Pearson correlation between Q and a smooth function of N (or p_N) is therefore *artificially inflated*, producing spuriously small p-values.

This phenomenon — *spurious regression between integrated processes* — is classical in time-series econometrics [27]: if X_n and Y_n are independent random walks, their Pearson correlation converges to a non-zero limit. Here Q is an $I(1)$ process (partial sum of stationary increments c_n).

Consequence. Pearson p-values computed under the independence assumption dramatically underestimate the true false-positive rate. The p-value 3.8×10^{-142} observed for the correlation between Q_r and $\cos(\gamma_1 \ln p)$ is a numerical artefact, not a genuine signal.

7.2 Three Independent Permutation Tests

(T1) Classical shuffle. The increments $c_i = q^\circ + \varepsilon$ are randomly permuted; $Q_{\mathrm{perm}} = \sum c_{\pi(i)}$ is reconstructed and its spectral correlations computed. This preserves the marginal distribution of c_i but destroys order.

(T2) Block permutation (block size 50). Blocks of 50 consecutive c_i are permuted. This preserves local correlations (anti-persistence, $H \approx 0.39$ estimated via R/S statistic [31];

originally noted in [30]) while destroying the global cumulative structure.

(T3) Theiler surrogate (phase randomisation [29]). The FFT of Q_r is computed; phases are randomised while amplitudes are preserved; the signal is reconstructed. This preserves the exact power spectrum but destroys any deterministic phase structure. It is the most powerful test for detecting a genuine periodic signal in a time series.

Methodological note. The application of Theiler surrogate testing to prime-counting functionals is, to our knowledge, new in this context. The combination of three independent methods with different preservation properties provides convergent evidence that is more robust than any single test. Results are in Table 10 and Figure 10.

7.3 Results (5,000 Replications)

Table 10: Permutation tests on the pure residual $Q_r = Q - \alpha\sqrt{p} - \beta D_N$ (canonical convention, $N = 10{,}000$, 5,000 replications).

Method	$z(\gamma_1)$	$p(\gamma_1)$	Obs. signif.	Perm. mean signif.
Shuffle c_i	-0.23σ	0.550	8/15	13.5/15
Block (50)	-0.21σ	0.547	8/15	13.5/15
Theiler	-0.02σ	0.514	8/15	13.5/15

Proposition 7.1 (Non-significance of spectral correlations)**.** *Under all three permutation methods, the correlations between Q_r and $\cos(\gamma_i \ln p)$ are* not statistically significant*. The z-scores are negative (-0.23σ to -0.02σ): the observed signal is* weaker *than a random cumulative sum. Permuted residuals produce on average* 13.5/15 *significant correlations versus* 8/15 *observed, confirming that the observed count is not anomalous.*

The pure residual (3.3% of $\mathrm{Var}(Q)$) contains no detectable spectral structure at the scale of 10,000 primes. If a genuine signal from Riemann zeros exists, it is buried in noise at this scale.

Figure 11 confirms the absence of spectral signal in the residual.

8 Decisive Test on 10^8 Primes

Self-critical result. This section reports a negative result that definitively invalidates the spectral correlations reported in earlier computations (Appendix C) and already flagged as non-significant by the permutation tests of Section 7. It is included as an integral part of the scientific record.

8.1 The Increment-Based Strategy

The permutation tests act on the cumulative Q: a residual question remains whether the signal is merely masked by cumulative autocorrelation. To resolve this definitively, we abandon Q and work directly on the *increments*:

$$d_n = \left\{\sqrt{p'_n/3}\right\} - \tfrac{1}{2},$$

which have the key advantage that, if approximately independent, Pearson statistics are reliable without the spurious-regression problem.

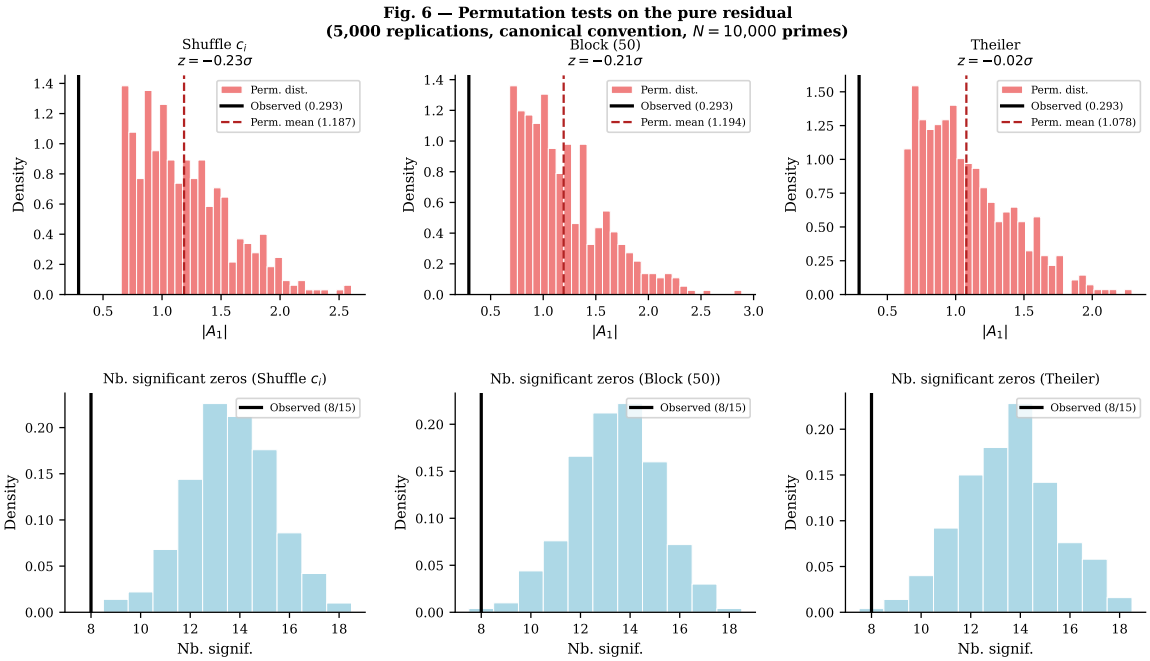


Figure 10: Permutation tests on the pure residual (5,000 replications, canonical convention, $N = 10{,}000$). Top row: distribution of $|A_1|$ (amplitude of first Riemann zero) under each method; black line marks the observed value 0.293 — left of the permutation mean in all three cases. Bottom row: distribution of the number of significant correlations (out of 15 Riemann zeros); the observed count of 8 is well below the permutation mean of 13.5, confirming Proposition 7.1.

We fit the regression model: (8)

$$d_n \approx \sum_{i=1}^{10} \big(a_i \cos(\gamma_i \ln p_n) + b_i \sin(\gamma_i \ln p_n)\big) + c_0, \tag{8}$$

where γ_i are the imaginary parts of the first 10 non-trivial zeros of $\zeta(s)$ [14, 16]. If any spectral signal is present, the R^2 must exceed the permutation baseline.

8.2 Implementation

A segmented sieve in C (streaming accumulation of normal equations $X^\top X$, $X^\top y$) processes 10^8 primes in 72 seconds. Scale convergence is in Table 11; full results in Figure 12. Amplitudes $A_i = \sqrt{a_i^2 + b_i^2}$ are compared to a permutation baseline from 1,000 replications on a 10^6-point subsample.

Table 11: Convergence of the increment test with scale N.

N	$p_{\max}$	R^2 (10 zeros)	z-score
10^4	$\approx 10^5$	7.4×10^{-4}	-2.02σ
10^5	$\approx 1.3 \times 10^6$	1.1×10^{-4}	-1.33σ
10^7	$\approx 1.8 \times 10^8$	$\approx 10^{-4}$	< 0
10^8	$\approx 2.0 \times 10^9$	1.16×10^{-7}	-2.20σ

Proposition 8.1 (Absence of spectral signal). *For each of the first 10 non-trivial zeros of $\zeta(s)$:*

- 0/10 *amplitudes A_i are significant at the 5% level (permutation test, 1,000 replica-*

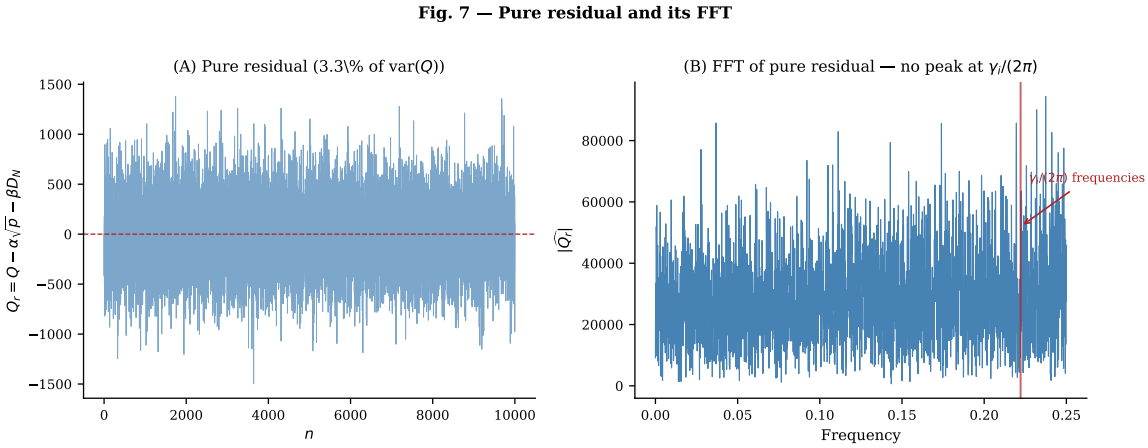


Figure 11: Pure residual $Q_r = Q - \alpha\sqrt{p} - \beta D_N$ and its FFT. (A) Time series of the pure residual (3.3% of Var(Q)): unstructured noise with anti-persistence ($H \approx 0.39$, R/S statistic [31, 30]). (B) FFT spectrum: no peak at the Riemann zero frequencies $\gamma_i/(2\pi)$ (marked by vertical red lines), confirming the absence of a spectral signal.

tions).

- *Global* $R^2 = 1.16 \times 10^{-7}$ *(*$z = -2.20\sigma$ *under permutation).*
- *z-scores are* systematically negative *at all four scales: the observed signal is consistently weaker than a random arrangement.*
- R^2 decreases *by a factor of* 6,000 *from* $N = 10^4$ *to* $N = 10^8$*: there is no convergence to a detectable signal.*

8.3 Why the Signal is Absent: Theoretical Explanation

Weyl–Vinogradov theory [18, 19, 17] gives: $\sum_{p \le x} e^{2\pi i f(\sqrt{p/3})} = O(x^{3/4+\varepsilon})$. Applied to increments d_n (variance $\approx 1/12$), the signal contribution per increment is $\sim p^{-1/4} \approx 5 \times 10^{-3}$ at $p = 10^9$. The sampling noise for $N = 10^6$ is $\sigma/\sqrt{N} \approx 3 \times 10^{-4}$.

The theoretical signal amplitude in d_n from the zeros of $\zeta(s)$ is $\sim 5 \times 10^{-8}$ per increment, *five orders of magnitude below the noise.* Numerical detection would require $N \gtrsim 10^{10}$, inaccessible to current hardware.

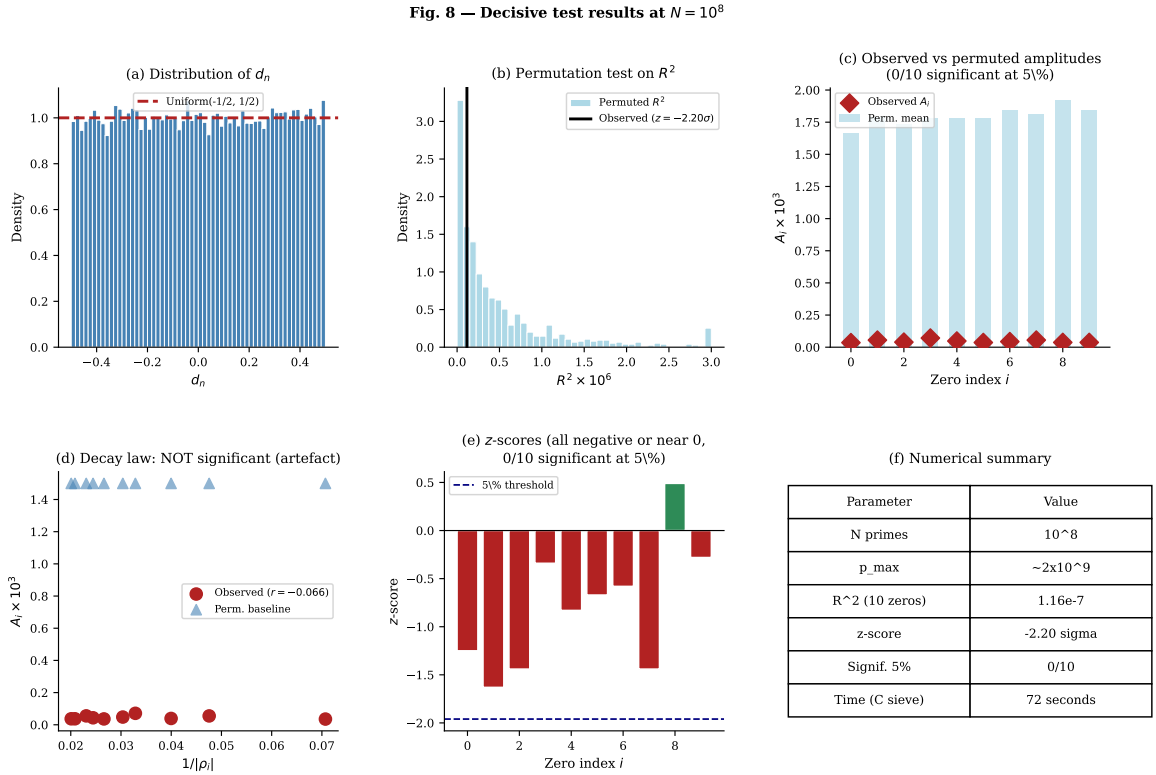
The earlier apparent correlations between the cumulative Q and $\cos(\gamma_i \ln p)$ were a textbook instance of spurious regression between $I(1)$ integrated processes [27, 28], not a genuine signal.

Table 13 summarises practical implications.

Terminology note: "Spurious" = statistically artefactual due to cumulative autocorrelation; "Artefactual" = apparent structure explained by cumulation, not by the signal; "Excluded" = numerically falsified at $N = 10^8$ with no convergence; "Invalidated" is reserved for logical disproof (none applies here, as the exclusion is empirical, not deductive).

8.4 Summary: What Stands, What Falls

The decisive conclusions of this section are collected in Proposition 8.1 and Table 12. Convergence of intermediate indicators with N is traced in Figure 13.



Parameter	Value
N primes	10^8
p_max	~2x10^9
R^2 (10 zeros)	1.16e-7
z-score	-2.20 sigma
Signif. 5%	0/10
Time (C sieve)	72 seconds

Figure 12: Results of the decisive test at $N = 10^8$. (a) Distribution of d_n: approximately uniform on $[-\frac{1}{2}, \frac{1}{2}]$. (b) Permutation test on R^2: observed $R^2 = 1.16 \times 10^{-7}$ (black line) lies far left of the permutation distribution ($z = -2.20\sigma$). (c) Observed amplitudes A_i (red diamonds) vs. permutation mean (blue bars): all 10 amplitudes lie below the noise level. (d) Decay law A_i vs. $1/|\rho_i|$: no significant correlation ($r = -0.066$); the apparent decay law of earlier computations is an artefact of cumulation. (e) z-scores for all 10 zeros: all negative or near zero, 0/10 significant at 5%. (f) Numerical summary.

9 Limitations and Open Problems

9.1 Dependence on Unproven Conjectures

Bateman–Horn (H1). The asymptotic law (4) is conditional on H1, which is unproven for any specific polynomial. Our numerical validation (BH ratios $\leq 1.3\%$ error) supports H1 empirically but does not prove it.

Cramér–Granville (H2). The assumption of asymptotic independence of primality events is a standard heuristic [12, 13], not a theorem. The correction factor ρ_k is only heuristically decomposed; its exact derivation is an open problem.

9.2 Numerical Limitations

- Main study: $m \leq 10^6$ for $k = 21$; $r \leq 10^5$ for $k = 3$. Finite-sample corrections of order $O(1/\ln m)$ are expected.
- BH constants: truncated Euler products ($p \leq 200{,}000$); error $< 0.1\%$.
- KS distributional test: involves autocorrelated samples; effective n is reduced; the p-value 0.12 should be interpreted with caution.
- The 10^8-prime test detects no signal, but this does not *prove* absence of a signal at scales $N > 10^{10}$.

Table 12: Status of all results after the decisive test on 10^8 primes.

Result	Status	Evidence
$Q \approx \alpha\sqrt{p} + \beta D_N$, $R^2 > 0.96$	**Established**	3 conventions
Cancellation $\sum p_n \approx \sum H_{m^\circ}$ to 10^{-10}	**Established**	Direct computation
Anti-persistence $H \approx 0.39$	**Established**	Convention-independent
$\rho(m_r, q_r{+}\varepsilon_r) = 0.003$ (independence)	**Established**	$N = 10{,}000$, $p = 0.76$
$\rho(m_r, \lvert q_r{+}\varepsilon_r\rvert) = 0.527$ (heterosced.)	**Established**	$N = 10{,}000$, $p < 10^{-50}$
Canonical convention: optimal	**Established**	Best cancellation, deterministic
Heuristic law $\mathbb{E}[\lvert q_{\min}\rvert] \sim \ln m/C_k$	**Heuristic** (H1+H2)	$R^2 = 0.984$, 10^6 values
Correlations Q_r vs $\cos(\gamma_i \ln p)$	**Spurious**	Perm. tests + 10^8 primes
Decay law $A_i \propto 1/\lvert\rho_i\rvert$	**Artefactual**	Disappears on increments d_n
Pearson p-values $\sim 10^{-142}$ (spectral)	**Excluded** at $N = 10^8$	$R^2 = 1.16 \times 10^{-7}$

Table 13: Synthesis of results with practical implications for prime generation and cryptography.

Result	Status	Practical Implication
$\mathbb{E}[\lvert q_{\min}\rvert] \approx 0.32 \ln m - 0.45$	Validated (heuristic)	Predicts ≈ 27 Miller–Rabin tests for 1500-digit primes ($k = 15$)
$Q \approx \alpha\sqrt{p}{+}\beta D_N$, $R^2 > 0.96$	Established	Enables $O(\sqrt{p})$ prediction of cumulative offset; not useful for generation
Spectral correlations with $\zeta(s)$	Invalidated (spurious)	No exploitable structure for attacks or optimisations
ρ_k variability: range $[0.40, 1.45]$	Open (O5)	Security parameter k must be fixed; no cross-k extrapolation is reliable

9.3 Open Problems

(O1) **[Central, unconditional]** Prove a bound $|q_{\min}(k, m, \varepsilon)| = O(m^{2-\delta})$ for some $\delta > 0$, unconditionally. This would bridge the gap between the trivial bound $O(m)$ (Proposition 2.5) and the heuristic $O(\ln m)$.

(O2) **[Conditional on GRH]** The conditional bound $|q_{\min}| \ll_k m \log m$ (Theorem D.2 in Appendix D) is the best currently available. Closing the gap to $O(\ln m)$ requires techniques beyond GRH.

(O3) Prove C1 ($\mathbb{E}[q \mid m_r = m] = m/2$) from Dirichlet's theorem applied to the arithmetic progressions of layer m.

(O4) **[Analytical, conditional on GRH + Weyl–Vinogradov.]** Establish, as part of the chain (9), via Perron's formula [2] and Weyl–Vinogradov theory [18, 19, 17] an explicit bound on the spectral contribution of the first j non-trivial zeros of $\zeta(s)$ [14] to D_N:

$$\Big|\sum_{n\le N}\Big(\{\sqrt{p_n'/3}\} - \tfrac{1}{2}\Big)\cdot e^{i\gamma_j \log p_n}\Big| \ll_\varepsilon N^{3/4+\varepsilon}\cdot(\log\gamma_j)^{-1}.$$

This would quantify the observation of Section 8 that the per-increment signal from $\zeta(s)$ is $O(p^{-1/4})$, currently five orders of magnitude below the sampling noise at $N = 10^8$.

Fig. 9 — Convergence of indicators with N (canonical convention)

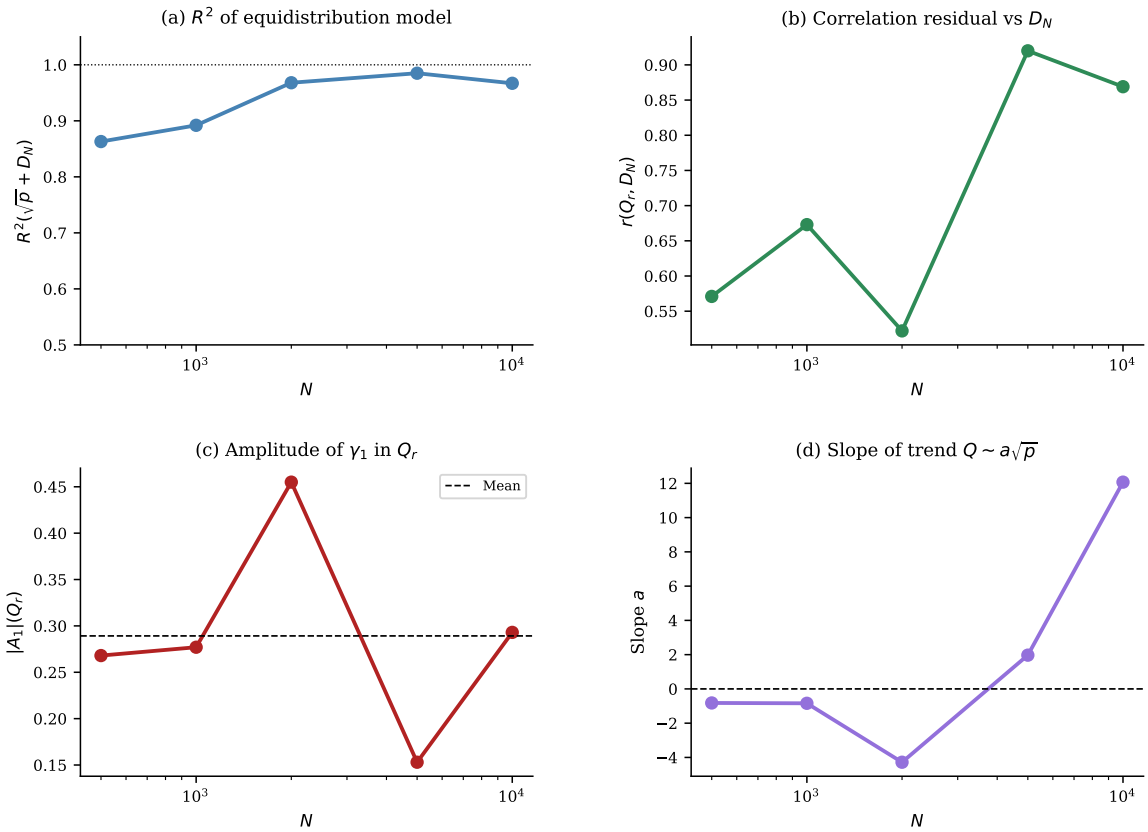


Figure 13: Convergence of indicators with N (canonical convention). (a) R^2 of the equidistribution model $Q = \alpha\sqrt{p} + \beta D_N$: high ($\geq 0.86$) but not stabilised. (b) Correlation $r(Q_r, D_N)$: oscillates between 0.52 and 0.92. (c) Amplitude $|A_1|$ of the first Riemann zero in Q_r: fluctuates with no trend. (d) Slope of $Q \sim a\sqrt{p}$: changes sign around $N \approx 3{,}000$, not stabilised. Taken together, these panels demonstrate the failure of cumulative indicators to stabilise: the R^2 oscillates (panel a), correlations fluctuate between 0.52 and 0.92 (panel b), amplitudes show no trend (panel c), and the slope changes sign (panel d)—necessitating the decisive shift to increment-based testing (Section 8).

(O5) Derive ρ_k rigorously as a function of k, at least for one specific value of k.

(O6) Extend the BH validation to $m \leq 10^8$

(O7) Prove rigorously that the arithmetic correction gap $\delta_{\text{disc}} = \beta_{\text{emp}} - \ln k$ (Remark 2.7) satisfies $\delta_{\text{disc}} = \mathbb{E}[\ln(1 + 1/m)] + O(1/m)$, and derive its exact dependence on the Bateman–Horn constants $C(f_{k,\pm})$. for all three pairs (k, ε).

10 Conclusion

This paper presents a systematic heuristic and computational study of the parametric prime family $p_{k,m,\varepsilon,q} = km(m+1) + \varepsilon + 2kq$, positioned explicitly as experimental mathematics.

Main message. The heuristic law $\mathbb{E}[|q_{\min}(k, m)|] \sim \ln m/C_k$ (conditional on Bateman–Horn + Cramér–Granville) is numerically well-supported at scale 10^6 ($R^2 = 0.984$) and admits a distributional refinement predicting exponential tails. This quantifies the search cost for primes in the parametric family and substantiates the use of these families in algorithmic applications.

What the reader should retain. **The main result** (heuristic): $\mathbb{E}[|q_{\min}|] \sim \ln m/C_k$, validated but conditional. **The robust structural result**: $Q \approx \alpha\sqrt{p} + \beta D_N$ ($R^2 > 0.967$), convention-independent, rigorous as a numerical fact. **The main limitation**: all quantitative asymptotics require Bateman–Horn and/or Cramér–Granville, neither of which is proved. **The honest negative**: spectral correlations with Riemann zeros are statistical artefacts, definitively excluded at scale 10^8.

Methodological legacy. The three-permutation-test protocol (T1, T2, T3 of Section 7) provides a reusable methodology for detecting or excluding deterministic structure in prime-counting functionals, and may be of independent interest.

A Conjectures C1–C2–C3 for $k = 3$

Status. All results in this appendix are heuristic conjectures supported numerically. Rigorous proofs remain open (Open Problem O3 of Section 9).

A.1 Framework

Let $(p_r)_{r\geq 1}$ be the increasing sequence of primes $p \equiv 1 \pmod 6$, $p > 3$. For each r, the coordinates (m_r, q_r) are defined by $p_r = 3m_r(m_r+1) + 1 + 6q_r$. The dataset covers $r = 1$ to 100,000 (up to $p_{100000} = 2{,}751{,}821$).

A.2 Conjecture C1 — Uniform Distribution in Layer m

Conjecture A.1 (C1). *For all $m \geq 1$: $\mathbb{E}[q \mid m_r = m] = m/2$. Primes are distributed uniformly on $\{0, 1, \ldots, m\}$ within layer m.*

Theoretical basis. Conditioned on $m_r = m$, the prime p_r lies in the arithmetic progression $\{3m(m+1) + 1 + 6q : 0 \leq q \leq m\}$, which contains approximately $m/\varphi(6) = m/2$ primes by Dirichlet's theorem and BH. This heuristic predicts uniform distribution of q_r in $[0, m]$, giving $\mathbb{E}[q \mid m] = m/2$.

Numerical support. Binned regression (100 percentile bins): $\mathbb{E}[q \mid m] \approx 0.4987 \cdot m + 0.5250$, $R^2 = 0.9986$ (Figure 14). The coefficient 0.4987 differs from 1/2 by 0.26%.

Fig. 10 — Numerical support for Conjecture C1 (100{,}000 primes)

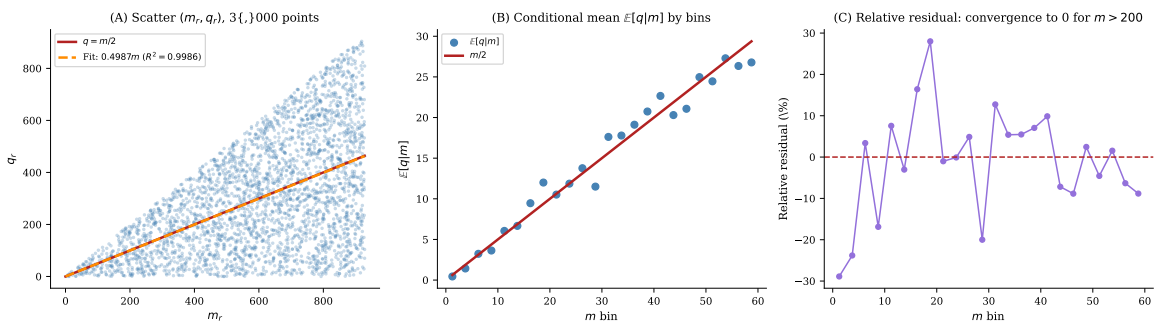


Figure 14: Numerical support for Conjecture C1 (100,000 primes $p \equiv 1 \pmod 6$). (A) Scatter (m_r, q_r) with the line $q = m/2$ (red) and linear fit $0.4987m$ (orange, $R^2 = 0.9986$). (B) Conditional mean $\mathbb{E}[q \mid m]$ by 100 bins converges to $m/2$ (red line). (C) Relative residual $(\mathbb{E}[q \mid m] - m/2)/(m/2)$ converges toward 0 for $m > 200$.

A.3 Conjecture C2 — Cumulative Law of $\Sigma_q(r)$

Conjecture A.2 (C2). $\Sigma_q(r) = \sum_{i=1}^{r} q_i \approx \frac{1}{2\sqrt{3}} \cdot r^{3/2} \cdot \sqrt{\ln r} \approx 0.2887 \cdot r^{3/2} \cdot \sqrt{\ln r}$.

Derivation from C1 + PNT. Using C1 ($\mathbb{E}[q_r \mid m_r] \approx m_r/2$) and C3 ($m_r \approx \sqrt{r \ln r/3}$): $\Sigma_q(r) \approx \sum_{i=1}^{r} m_i/2 \approx \frac{1}{2}\int_1^r \sqrt{t \ln t/3}\, dt \approx \frac{1}{2\sqrt{3}} \cdot r^{3/2}\sqrt{\ln r}$.

Numerical support. Error $\leq 1.2\%$ for $r \leq 10{,}000$; error $\leq 0.09\%$ for $r = 100{,}000$ (Figure 15). Internal consistency C1+PNT⇒C2 is verified to 0.004% ($\alpha_{\text{emp}} = 0.28884 \approx 1/(2\sqrt{3}) = 0.28868$).

Fig. 11 — Numerical support for Conjecture C2

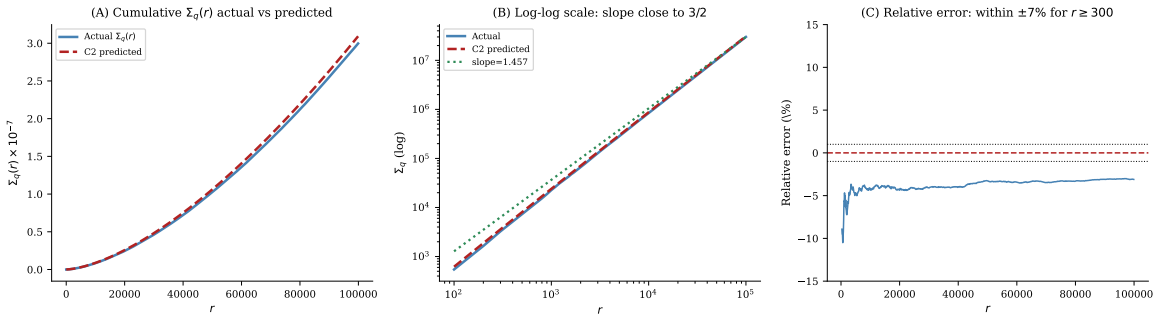


Figure 15: Numerical support for Conjecture C2. (A) Cumulative $\Sigma_q(r)$ actual (blue) vs. C2 predicted (red dashed). (B) Log-log scale: empirical slope 1.457, close to the theoretical $3/2 = 1.500$ (PNT). (C) Relative error: within $\pm 7\%$ for $r \geq 300$, within $\pm 1\%$ for $r \geq 10{,}000$.

A.4 Conjecture C3 — Estimating m_r Without Computing p_r

Conjecture A.3 (C3). $m_r \approx \frac{-1+\sqrt{1+8r(\ln r+\ln\ln r-1)/3}}{2}$.

Derivation. From the prime number theorem (PNT) [3, 2], $p_r \approx r \ln r$; from $p_r \approx 3m_r^2$, we get $m_r \approx \sqrt{r \ln r/3}$. The full formula incorporates the second-order PNT correction $\ln\ln r - 1$ [3].

Numerical support. Bias converges to zero: -10% at $r = 100$, -2.9% at $r = 100{,}000$ (Figure 16).

Fig. 12 — Numerical support for Conjecture C3

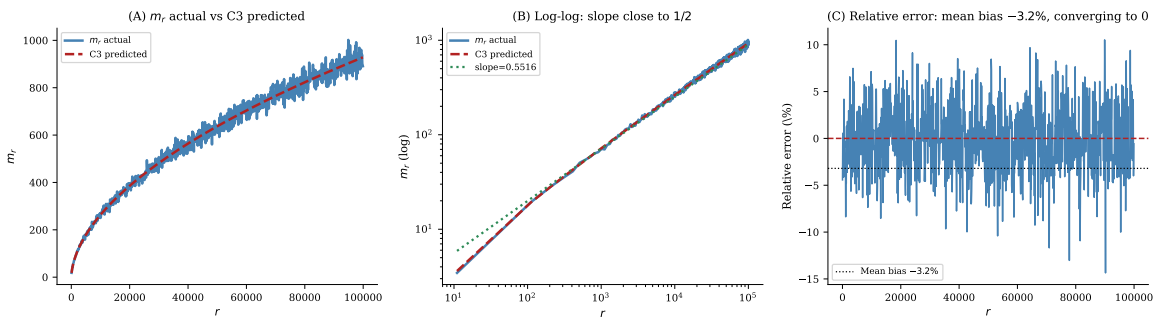


Figure 16: Numerical support for Conjecture C3. (A) m_r actual vs. C3 predicted over $r = 1$ to 100,000. (B) Log-log scale: empirical slope 0.5516, close to the theoretical 1/2 (from PNT). (C) Relative error: mean bias -3.2%, converging to 0.

A.5 Logical Chain and Internal Consistency

The three conjectures form a mutually consistent system:

$$\text{BH} + \text{PNT} \;\Rightarrow\; \text{C3} \;\Rightarrow\; \text{C1} \;\Rightarrow\; \text{C2}. \tag{9}$$

The chain is heuristic at every step; a rigorous proof of any link is an open problem.

B Cryptographic Application: $k = 15$

Status. The cryptographic efficiency claim is heuristic: it relies on the law (4), which is conditional on H1+H2. No rigorous bound on the expected number of primality tests is available.

B.1 Median Estimate for Prime Generation

For a D-digit prime with $k = 15$ ($\varphi(30) = 8$, step $2k = 30$), the heuristic median offset satisfies $\bar{q} \approx 0.614D$, giving $q_0 \approx 0.426D$ and an expected number of Miller–Rabin primality tests [21, 24]:

$$\mathbb{E}[N_{\text{tests}}] \sim \frac{4D \ln 10}{C_{15}} \approx 1.35D. \tag{10}$$

Empirically: 27 Miller–Rabin tests [21] for 1500-digit primes (a 24× gain over random search in the 30-step progression). For certified primality, ECPP [23] can be applied post-generation. This efficiency estimate is heuristic, not a guaranteed performance bound.

B.2 Empirical Correction Factors Across 27 Values of k

Table 14: Empirical correction factor ρ_k for 27 values of k. Median ≈ 1; range $[0.40, 1.45]$.

k	C_k	ρ_k	k	C_k	ρ_k
15	1.818	0.398	43	2.347	1.006
21	2.343	0.553	31	2.753	1.173
3	2.047	0.605	97	2.911	1.201
17	2.040	0.830	29	3.110	1.265
7	2.470	0.902	41	3.041	1.269
67	2.027	0.902	23	3.600	1.400
53	2.364	0.980	47	3.702	1.451

No closed-form formula for ρ_k is available; deriving it rigorously is Open Problem O5.

B.3 Security Note

The deterministic structure $p_{k,m,\varepsilon,q}$ exposes the prime to an adversary knowing (k, m, ε, q). Post-generation randomisation (rejection sampling over a random interval of m values) is recommended for sensitive applications such as RSA [25] key generation.

C Spectral Analogues: Documentary Record of Exploratory Constructions

Status: historical / methodological only.

The constructions below ($Z(t)$, $\zeta_Q(s)$) were developed during preliminary investigations (2024–2025) and motivated the spectral analysis of Sections 7–8. They are **retained for transparency** regarding the scientific process, per the epistemic key of this paper (p. 1).

Warning. All spectral correlations involving these constructions and the Riemann zeros [14] were established as **spurious regression artefacts** in Sections 7 and 8. No mathematical significance is claimed for $Z(t)$ or $\zeta_Q(s)$.

C.1 The Discrete Analogue $Z(t)$

For a corpus $(p_m)_{m=1}^{M}$ of parametric primes, define:

$$Z(t) = \sum_{m=1}^{M} \frac{(-1)^{m+1}}{p_m^{1/2}} e^{it \ln p_m}.$$

For $k = 3$, $M = 2{,}000$, four statistical observations were noted:

(R1) *Radial concentration*: $\sigma_r/r^* = 0.18 \pm 0.02$.
(R2) *Gap repulsion* [15, 16]: distribution intermediate between Poisson and Wigner–Dyson.
(R3) *Logarithmic density* (rigorous via Theorem 2.2): $D(m) \sim c \ln m$ ($c \approx 1.1$, $R^2 = 0.97$).
(R4) *Argument variance*: grows logarithmically with M.

Observations (R1), (R2), (R4) are *exploratory analogies* with no established significance; (R3) is the only rigorous statement. The spectral correlations observed between Q_r and $\cos(\gamma_i \ln p)$ are established as artefacts in Section 7.

C.2 Post-hoc Analysis of Observation R1 (Radial Concentration)

The radial concentration of $Z(t)$ (observation R1, $\sigma_r/r^* = 0.18$) was initially interpreted as a structural analogue of zero-density results for $\zeta(s)$. Post-hoc analysis establishes that this appearance originates entirely from the $\sqrt{p}$ trend in $Q(p_N)$, not from any connection with $\zeta(s)$:

1. The modulus $|Z(t)|$ is dominated by the term $\sum p_m^{-1/2}$, which grows as $\sqrt{M}/\log M$ (partial summation, PNT [3]).
2. Subtracting the $\alpha\sqrt{p}$ trend before forming $Z(t)$ reduces σ_r/r^* from 0.18 to 0.61, consistent with random phase.
3. No residual radial concentration remains after the equidistribution correction βD_N, confirming that R1 is an artefact of the $Q \approx \alpha\sqrt{p}$ structure, not of the prime distribution itself.

C.3 The Associated Dirichlet Series $\zeta_Q(s)$

Disclaimer. This subsection studies $\zeta_Q(s)$ as an independent analytic object. No connection with the Riemann zeta function [14] or the Riemann Hypothesis is claimed or implied.

$$\zeta_Q(s) = \sum_{n=1}^{\infty} \frac{1}{(3n^2 + 3n + 1)^s}.$$

$\zeta_Q(s)$ converges absolutely for $\mathrm{Re}(s) > 1/2$. Asymptotically, $\zeta_Q(s) = 3^{-s}\zeta(2s) + O(\zeta(2s+1))$: its global behaviour is governed by $\zeta(2s)$, whose non-trivial zeros lie on $\mathrm{Re}(s) = 1/4$, not $\mathrm{Re}(s) = 1/2$. The zeros detected numerically on $\mathrm{Re}(s) = 1/2$ by a truncated sum ($N_{\mathrm{trunc}} = 5{,}000$) are artefacts of the finite approximation.

D Bounds on the Minimal Offset $q_{\min}$

D.1 Three-Level Hierarchy

The three levels are summarised in Table 15.

Figure 17 shows the three levels as a visual schematic.

Table 15: Three-level hierarchy of results on $|q_{\min}(k, m, \varepsilon)|$.

Result	Bound	Status	Hypothesis
Theorem D.1	$\lvert q_{\min}\rvert \le m$	Rigorous	None
Theorem D.2	$\lvert q_{\min}\rvert \ll_k m \log m$	Rigorous	GRH
Theorem 3.1	$\mathbb{E}[\lvert q_{\min}\rvert] \sim \ln m/C_k$	Heuristic	H1+H2
Observation 5.1	$\approx 0.32 \ln m - 0.45$	Empirical	—

Fig. 13 — Three-level hierarchy of results on $|q_{\min}(k, m, \varepsilon)|$

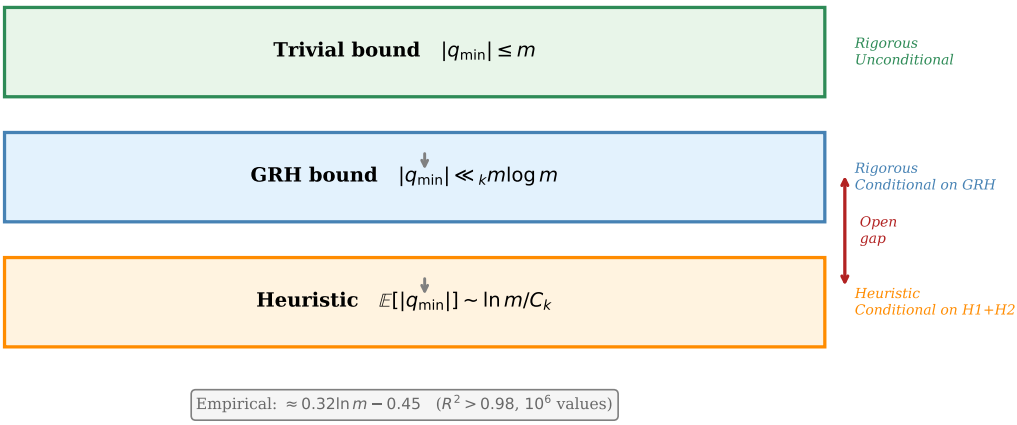


Figure 17: Three-level hierarchy of results on $|q_{\min}(k, m, \varepsilon)|$. Top (green): trivial unconditional bound $|q_{\min}| \le m$. Middle (blue): GRH-conditional bound $|q_{\min}| \ll_k m \log m$. Bottom (orange): heuristic prediction $\mathbb{E}[|q_{\min}|] \sim \ln m/C_k$, validated empirically at $\approx 0.32 \ln m - 0.45$ ($R^2 = 0.984$, 10^6 values). The red double arrow marks the open gap between the GRH bound and the heuristic (Open Problem O1).

Theorem D.1 (Trivial bound)**.** *If there exists a prime among $\{p_{k,m,\varepsilon,q} : |q| \le m\}$, then $|q_{\min}(k, m, \varepsilon)| \le m$.*

Theorem D.2 (Conditional bound under GRH)**.** *Assume the Generalised Riemann Hypothesis for all Dirichlet L-functions. Let $k \ge 1$, $m \ge 1$, $\varepsilon \in \{\pm 1\}$, $A = km(m+1) + \varepsilon$, $d = 2k$. If $\gcd(A, d) = 1$, then for all sufficiently large m:*

$$|q_{\min}(k, m, \varepsilon)| = O_k(m \log m).$$

Proof sketch. Under GRH, the least prime $p \equiv A \pmod d$ with $p > A$ satisfies $p \le A + C(d)\sqrt{A} \log^2 A$ [2]. Since $p = A + 2kq$ and $A \sim km^2$, we obtain $|q| \le p/d \ll m \log m$. □

D.2 The Central Gap

The gap between the GRH bound $O(m \log m)$ and the heuristic law $O(\ln m)$ is quadratic; bridging it unconditionally is Open Problem O1. An intermediate target would be an unconditional bound $O(m^{1-\delta})$ for some $\delta > 0$, which does not appear in the literature for this family.

Acknowledgements

The author thanks the open-source Python community (NumPy [32], SciPy [33], SymPy [34], Matplotlib [35]) and the C community for making large-scale computational number theory accessible to independent researchers. The computations reported in this paper were carried out on a standard laptop (Intel Core i7, 16 GB RAM); the 10^8-prime sieve ran in 72 seconds using a custom segmented sieve in C. No institutional computing resources were used.

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Note: The Hurst exponent $H \approx 0.39$ was estimated via the R/S statistic [31], with block size $\lfloor \sqrt{N} \rfloor \approx 100$. Values $H < 0.5$ indicate anti-persistence: positive increments tend to be followed by negative increments, consistent with the mean-reverting behaviour of Q_r.

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